COURSE: BUSINESS INTELLIGENCE (IT730A)

DATA DRIVEN CULTURE

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Agenda

Data Driven Culture

1. What is that?
2. Problems
3. Opportunities

What is Data Driven Culture?

Organizations that consistently collect, analyze, and act on data.

A mindset that sees data as critical for success, not an optional resource.

- Teams that naturally ask analytics-driven questions: **“What does the data tell us?”**
- A shared culture where data and analytics **guide decisions at all levels**—executives, managers, and frontline staff.
- **Leadership commitment:** especially to Big Data Analytics and AI initiatives.

Distinction (Analytics & Group Development

	Dependency & Inclusion	Counter-Dependency & Fight	Trust & Structure	Work & Productivity
Descriptive Analytics				
Predictive Analytics				
Prescriptive Analytics				
No analytics				

- **Territorial analytics (white area):** team that is in the early stages of group development but frequently uses analytics but findings are seldom openly shared with other members.
- **No analytics:** team that evolves as a group and becomes successful without frequently using analytics.
- **Data-driven culture:** team that frequently uses analytics is when making decisions. All members have a brief understanding of what is meant by business intelligence & analytics and data-driven organizations.

Data Driven Culture in the AI Era

AI/analytics implementation is **not a linear pipeline** ending at deployment.

AI requires **continuous learning and development** cycles.

Success also depends on **business change management**, not just model accuracy.

Trust in automated decisions (by employees and customers) is **essential.**

Implementation in Practice

1. Deep understanding of the business problem and observing the real-world process.
2. Awareness of model failure modes and their implications.
3. Trust-building with managers and stakeholders.
4. Familiarity with the technology stack—where the model will live.
5. Accountability for changes and outcomes.
6. Monitoring performance post-deployment.

How to be successful

- Build **multidisciplinary data science teams** with complementary expertise.
- During hiring, ask about **deployment experience**, not only modeling skills.
- Establish a **deployment pipeline** for every project.
- Secure executive buy-in to **ensure adoption**.
- Promote data literacy, data usage incentives, and analytics-driven decision making.
- **Depersonalize** decision-making
- **Experience Analysis** of customers

It is estimated that roughly a third of the organizations make a successful shift and become data-driven...

Why?

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The Deployment Challenge

Despite massive investment in big data, **deployment remains rare**.

Statistics:

- Only **13%** of data scientists report that their **models are always deployed**.
- AI usage: **91.5%** of firms **invest in AI**, but only **14.6% deploy** it widely.
- Only **1 in 10 companies** see significant **financial benefits**.

Data often breaks at deployment: unreliable, changing, or inconsistent with training data.

Non-Technical Barriers

- **Lack** of management **support**
- Skill shortages
- **Resistance** among employees (“the model is replacing my judgment”)
- Poor data quality
- Difficulty establishing a culture where people **openly discuss insights**
- Existing maturity models for analytics don't address group development
- Difficult to **change the whole company culture** especially in big companies

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Why strive for it?

Industry leaders (Google, Amazon) excel by leveraging data-centric business models.

- Better measurement → **better management**.
- **Improved forecasting**, decision quality, and interventions.
- AI can **automate tasks** previously manual or impossible.
- A way to **find out your problems**

Competitive Advantage

- Analytics-driven organizations often become **top performers**.
- Predictive analytics **improves productivity** in operational settings.
- Frequent use of analytics **improves group dynamics**—less anecdote, more evidence.
- Investing in group development prevents territorial analytics.

Conclusion

Data Driven Culture

1. A data-driven culture is built on behaviors, not technology.

2. Deployment—not model building—is the hardest step.
Most challenges are organizational, not technical.

3. The rewards are substantial: better decisions, higher productivity, and sustainable competitive advantage.

Thank you very much!

QUESTIONS?

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